

Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 6: Variation or Copies? Debating the Trade-offs of Sexual and Asexual Reproduction in Kenyan Crops

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	B	C	D	C	A	A

USE AFTER: PREDICT PHASE — Predict: two-farm scenario (Farm A cassava cuttings, Farm B maize seeds) near Kakamega

Q1. A new fungal disease reaches two farms near Kakamega. Farm A has cassava grown from cuttings taken from one parent plant. Farm B has maize grown from seeds collected from many different plants. What is the most likely result?

- A. Farm A is likely to be badly affected because all its plants are identical, while on Farm B some plants may survive because they differ**
- B. Farm A is safe because plants grown from cuttings are stronger than plants grown from seeds
- C. Both farms are affected in the same way because a disease attacks every plant equally
- D. Farm B loses every plant because seed-grown plants are always weaker

Correct answer: A. Cuttings give genetically identical plants, so if one is susceptible all are. Seed-grown maize shows variation, so some plants may resist the disease.

USE AFTER: OBSERVE PHASE — Observe: case card (1) Banana wilt in Kisii

Q2. In Kisii, banana suckers were taken from one infected plant and planted across a shamba. Banana wilt then spread through almost the whole shamba. Why did it spread so easily?

- A. Bananas must be cross-pollinated, and the pollinators carried the disease
- B. Suckers are genetically identical to the parent, so all the plants are equally susceptible to the disease**
- C. The suckers grew from seeds that carried the disease
- D. Suckers show a lot of variation, so the disease could change to attack each plant

Correct answer: B. Suckers are produced by asexual reproduction, so they are clones of the parent. A disease that can infect one plant can infect all of them.

USE AFTER: OBSERVE PHASE — Observe: case card (2) Maize seed variation in a drought-hit field in Machakos

Q3. During a drought in a field in Machakos, some seed-grown maize plants survived while many others died. What best explains why some plants survived?

- A. All the maize plants were identical, so the survivors were just lucky
- B. Maize reproduces asexually, so the survivors were clones of a drought-resistant parent
- C. Sexual reproduction produced seeds with different gene combinations, so some plants could tolerate drought**
- D. The drought changed the genes of every plant so that they could survive

Correct answer: C. Meiosis and fertilisation mix genes, so seed-grown plants differ from each other. Some of these differences can help a plant survive when conditions change.

USE AFTER: OBSERVE PHASE — Observe: case card (3) Cassava cuttings in Busia

Q4. A farmer in Busia has one cassava plant with a high yield and a sweet taste. Why is it a good choice to plant cuttings from this plant?

- A. The cuttings will give new varieties with different tastes, so the farmer can choose the best one
- B. The cuttings need another cassava variety nearby for pollination before they can grow
- C. The cuttings will be resistant to every disease because they come from a healthy plant
- D. The cuttings quickly and cheaply give many plants that are identical to the parent, with the same high yield and taste**

Correct answer: D. Asexual reproduction gives fast, cheap, identical copies (clones) without a partner. This keeps the useful features of the parent.

USE AFTER: EXPLAIN PHASE — Explain: 'why a stem cutting grows with no flower while a lone avocado tree cannot set fruit'

Q5. A single avocado tree in a compound is covered in flowers every season but never sets fruit. No other avocado tree grows nearby. What is the best explanation?

- A. Flowers cannot form fruit unless the plant is grown from seed
- B. The tree has too many flowers and uses up all its energy before fruits can form
- C. Fruit and seeds need pollination and fertilisation, and the avocado often needs pollen from another tree, which is not available here**
- D. Avocado reproduces only by cuttings, so flowers cannot make fruit

Correct answer: C. Fruits develop after pollination and fertilisation. Many avocado and pawpaw trees need cross-pollination, so a lone tree may flower but not fruit.

USE AFTER: EXPLAIN PHASE — Explain: structured debate on 'A Kenyan farmer should plant only cuttings, never seeds'

Q6. In the class debate on the motion 'A Kenyan farmer should plant only cuttings, never seeds', which is the strongest point for the side that opposes the motion?

- A. Seeds give variation, so some plants may survive a disease or drought, and seeds can also spread to new areas**
- B. Cuttings always need a partner plant for pollination, but seeds do not
- C. Seeds give plants that are identical to the parent, so the farmer keeps the same yield
- D. Cuttings are slower than seeds, so farmers lose time

Correct answer: A. Seeds come from sexual reproduction, which gives variation and helps dispersal. Variation means a single disease or drought is less likely to destroy all the plants. The other choices are wrong because cuttings are fast, need no partner, and seeds do not give identical offspring.

USE AFTER: END OF LESSON — End of lesson: revised class model and exit ticket on advising a farmer about cassava and beans

Q7. Which statement is the best conclusion about sexual and asexual reproduction in plants?

- A. Neither is better in all conditions: asexual gives fast identical copies, sexual gives variation, and the best choice depends on the crop and the environment**
- B. Both give identical offspring, so the choice does not matter to the farmer
- C. Sexual reproduction is always better because it gives variation
- D. Asexual reproduction is always better because it is faster and needs no partner

Correct answer: A. Cloning is useful for quickly copying a good variety but is risky if a disease arrives. Variation from seeds helps survival when conditions change. The better strategy depends on the situation.